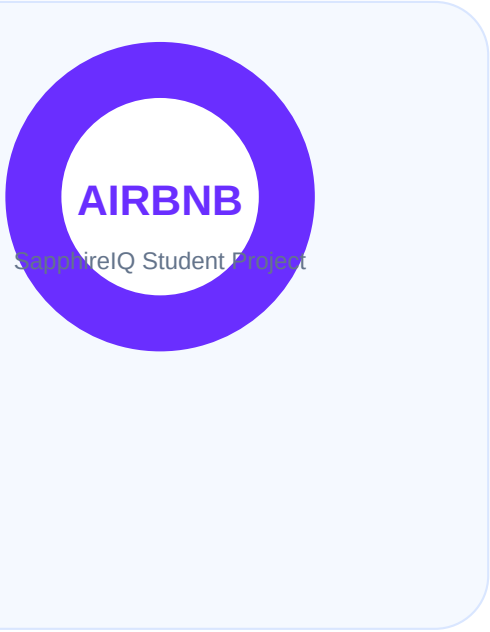
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FULL STACK DEVELOPMENT • ADVANCED STUDENT PROJECT

Airbnb Clone (Full Stack)

React + Node.js/Express + MongoDB/PostgreSQL + JWT + Maps

A complete listings and booking platform, covering the backend foundation, connected search UI, end-to-end booking, maps/reviews/dashboard, and deployment.



5	Full	Maps	Booking
Modules	Stack	Integration	Core Flow

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01 / PROJECT OVERVIEW

Airbnb Clone (Full Stack)

An advanced full stack capstone project modeling a complete listings-and-booking platform.

Project Description

This curriculum provides a step-by-step blueprint for building the Airbnb Clone as a full stack capstone, covering the backend data model and CRUD APIs, a connected listings/search frontend, an end-to-end booking flow, maps/reviews/host dashboard features, and final deployment and polish.

Project Scope

Build the backend foundation, connect the frontend to live listings/search, wire up the full booking flow, add maps/reviews/dashboard, and deploy the complete platform.

Prerequisites

- Prior Projects: At least one completed frontend and one completed backend project.
- Full Stack Basics: Comfortable connecting a React app to a self-built API.
- Tooling: Node.js (v18+), a database, and a Maps API key.

Target audience: Advanced students ready for a multi-week full stack capstone project.

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02 / PROJECT ROADMAP

Five-Module Airbnb Clone (Full Stack) Roadmap

Progress through the project in a structured sequence, from core foundations to the completed

1	**Backend Foundation**	Data Model & Core APIs	Schema design, REST API design, auth middleware
2	**Frontend: Listings & Search**	Connected Search UI	API integration, filter UI, responsive grid layouts
3	**Booking Flow End-to-End**	Connecting Complex Logic	Date-picker integration, overlap validation, transactional writes
4	**Maps, Reviews & Host Dashboard**	Feature Completeness	Maps API, review aggregation, role-based UI
5	**Deployment & Polish**	Shipping a Capstone	Deployment, environment separation, performance basics

Student Workflow

Complete each module in sequence, run both frontend and backend after every milestone, and use the hands-on exercises to extend the implementation.

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01 / MODULE 1

Backend Foundation

Build the project step by step and apply each concept through the hands-on exercise.

Learning Objectives

- Build the full data model for listings, bookings, and reviews.
- Implement CRUD APIs for each entity.
- Protect sensitive routes with auth middleware.

Core Concepts

- Full Data Model: designing every core entity before frontend work begins.
- Auth Middleware: gating write operations behind authentication.

Tools & Technologies

- Express
- Mongoose/Sequelize
- JWT

Hands-on Student Exercises

- Backend supports all core Airbnb entities

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02 / MODULE 2

Frontend: Listings & Search

Build the project step by step and apply each concept through the hands-on exercise.

Learning Objectives

- Build the listing grid connected to the live API.
- Implement search and filter UI.
- Handle loading and empty states.

Core Concepts

- Live API Integration: replacing mock data with real backend responses.
- Responsive Grid Layouts: presenting many listings cleanly.

Tools & Technologies

- Axios/fetch
- Responsive CSS

Hands-on Student Exercises

- Users can browse and filter real listings

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03 / MODULE 3

Booking Flow End-to-End

Build the project step by step and apply each concept through the hands-on exercise.

Learning Objectives

- Connect the booking UI to backend availability logic.
- Validate date selection against existing bookings.
- Confirm successful bookings without conflicts.

Core Concepts

- End-to-End Booking Flow: coordinating frontend and backend validation.
- Overlap Validation: preventing double-bookings from the UI down to the database.

Tools & Technologies

- Date-picker library
- Booking API

Hands-on Student Exercises

- Users can complete a booking without conflicts

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04 / MODULE 4

Maps, Reviews & Host Dashboard

Build the project step by step and apply each concept through the hands-on exercise.

Learning Objectives

- Add a map view of listings.
- Build a review system for completed stays.
- Build a host dashboard for managing listings.

Core Concepts

- Review Aggregation: computing average ratings from individual reviews.
- Role-Based UI: showing different views to hosts vs. guests.

Tools & Technologies

- Maps API
- Role-based components

Hands-on Student Exercises

- Hosts can manage listings; guests can review stays

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05 / MODULE 5

Deployment & Polish

Build the project step by step and apply each concept through the hands-on exercise.

Learning Objectives

- Deploy both frontend and backend.
- Separate development and production environment configs.
- Add image optimization and final UX polish.

Core Concepts

- Environment Separation: distinct configs for dev vs. production.
- Performance Basics: optimizing images and load times before launch.

Tools & Technologies

- Render/Vercel/Netlify
- Image optimization

Hands-on Student Exercises

- Fully live, end-to-end Airbnb-style platform

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PROJECT OUTCOME

What You Will Build

Application Structure

React Frontend

Search, booking, dashboard

Express API

Listings, bookings, reviews

Maps API

Location & discovery

Skills Practiced

- Full data modeling for a multi-entity platform.
- Connecting search/filter UI to a live backend.
- End-to-end booking logic with conflict prevention.
- Maps integration and role-based dashboards.
- Full stack deployment and performance polish.

Use the module exercises as checkpoints while building your application.